MILITARY SPECIFICATION

CLOTH, COATED AND LAMINATED, CHEMICAL PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document is for one type of polyethylene coated and laminated cloth. It is to be used for the manufacture of a ground sheet for a chemical protective patient wrap (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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MILITARY

MIL-P-44305 - Plastic Film, Flexible, Polyamide and Polyethylene

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-601 - Rubber: Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-282 - Filter Units, Protective Clothing Gas Mask Components and Related Products, Performance Test Methods
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

CRDC-SP-84010 - Laboratory Methods for Evaluating Protective Clothing Systems Against Chemical Agents

(Copies may be obtained from the Chemical Research and Development Center, ATTN: DRSMC-CLW-P(A), Aberdeen Proving Ground, MD 21010.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Standard sample. The coated cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to the characteristics for which the standard sample is referenced (see 6.3).

3.3 Materials.

3.3.1 Base fabric. The base fabric shall be plain woven using the yarn specified in 3.3.2. The weight of the base fabric shall be 3.0 ± 0.3 ounces per square yard. The construction of the woven fabric shall be 10 by 10 yarns per inch (± 1/2 yarn per inch).

3.3.2 Yarn. The yarn shall be made from slit film ribbons of high density polyethylene polymer, shall be 950 ± 50 denier, and shall be suitably compounded, oriented, and heat stabilized to meet the requirements of the finished coated and laminated cloth.

3.3.3 Coating compound. The coating compound shall be a low density polyethylene coating suitably compounded and pigmented such that the finished, coated and laminated cloth conforms to the requirements of 3.5 and 3.6. The coating compound shall contain a flame inhibitor.

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3.3.4 Coated cloth. The base fabric of 3.3.1 shall be coated on each side with 1.4 ± 0.1 ounces per square yard of the coating compound specified in 3.3.3.

3.3.5 Plastic film. The plastic film shall be a flexible, polyamide and polyethylene plastic film conforming to MIL-P-44305.

3.4 Coated and laminated cloth. The back or polyethylene side of the plastic film specified in 3.3.5 shall be laminated to one side of the coated cloth specified in 3.3.4. The plastic film side of the coated and laminated cloth shall be the face side.

3.5 Physical requirements. The finished coated and laminated cloth shall conform to the requirements specified in table I when tested as specified in 4.4.3.

TABLE I. Coated and laminated cloth requirements

Characteristic	Requirement	
	Min.	Max.
Weight, oz/yd^2	6.5	8.0
Breaking strength, pounds:		
Initial:		
Warp	135	
Filling	135	
After oven aging:		
Warp	130 1/	
Filling	130 1/	
Resistance to abrasion, percent strength loss:		
Warp	--	20
Filling	--	20
Tearing strength, grams:		
Warp	30000	--
Filling	30000	--
Resistance to leakage:		
Initial	Pass	--
After cold crack	No leakage 2/	--
Resistance to blocking, rating	--	No. 2
Resistance to toxicological agents, minutes:		
Mustard (HD)	300	--
Thickened GD	300	--

1/ No blistering or delamination of the coated and laminated cloth shall be evident.

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2/ Leakage is defined as the appearance of one or more of water droplets in the 4-1/2 inch test area.

3.6 Color. The color of both sides of the coated and laminated cloth shall be dark green.

3.6.1 Matching. The color of both sides shall match the standard sample (see 6.3), when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles. The color shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.7 Width. The minimum overall width of the finished coated and laminated cloth, after selvage trimming, shall be 45 inches. Selvages shall be trimmed to give a straight, uniform edge. If the plastic film used is narrower than the 45 inches overall width requirement, it shall be positioned so that the edges are equidistant from the trimmed edges of the cloth.

3.8 Length and put-up. Each roll shall contain no less than 180 yards nor more than 220 yards. The maximum number of pieces per roll shall be four, and no single piece shall be less than 20 yards in length. The ends of the pieces shall be overlapped and not joined by a seam. The face side of the finished, coated and laminated cloth shall be wound toward the outside of the roll. The cloth shall be put up in rolls as specified in 5.1.

3.9 Workmanship. The finished cloth shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the appearance and length defects listed in table III and 4.4.2.2 and tested for the characteristics specified in table IV. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item examination.

4.4.2.1 Yard-by yard examination. The coated and laminated cloth shall be examined for the defects listed in table III. The required yardage of each roll shall be examined on one side only, however, the side shall be alternated so that alternate rolls are examined on the face and back, respectively. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs The lot size shall be expressed in units of yards. The sample unit shall be 1 linear yard. The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. The number of rolls from which the sample yardage is to be selected shall be as specified in table II. The sample yardage shall be apportioned equally among the selected rolls.

TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number 2/
Up to 1,200 inclusive 1/	3	0
1,201 to 3,200 inclusive	5	0
3,201 to 10,000 inclusive	8	0
10,001 to 35,000 inclusive	13	0
35,001 to 150,000 inclusive	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

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2/ Applicable to length examination only (see 4.4.2.2).

TABLE III. Yard-by-yard examination defects

Defect	Classification Major	Minor
Any cut, hole, or tear	X	
Any abrasion mark or scratch that extends through the coating to a point where base cloth yarns or cloth weave can be seen	X	
Any pinhole 1/	X	
Any uncoated area	X	
Any pit, blister, tunnel, or delamination of coating	X	
Any lump or embedded foreign matter that protrudes so as to be readily abraded from material or easily removed	X	
Crease or wrinkle resulting in adhesion of surface or delamination of coating when corrected by manual pressure	X	
Any spot, stain, or streak, more than 1 inch in combined direction that cannot be readily removed with naptha or water 2/	X	
Spliced cloth (base fabric or laminated cloth) or impression in the film or coating from splicing threads	X	
Width of laminated cloth or plastic film less than minimum specified		X
Color not as specified		X
Coating improperly applied	X	
Plastic film improperly applied	X	
Ripples, waviness, or dimensional distortion		X
Edges not straight and uniform, or edges folded, rolled, or scalloped		X
Any objectionable odor 3/		X

1/ The examination for pinholes shall be performed in accordance with the through-light inspection procedure in MIL-STD-1487.

2/ Clearly visible at normal inspection distance (approximately 3 feet).

3/ Odors of chemical commonly used in coating compounds shall not be regarded as objectionable.

4.4.2.2 Length examination. Each roll of cloth used in the yard-by-yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table II or if the total of the actual lengths of the sample rolls is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

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Total length of roll less than 180 yards or more than 220 yards.
Total length of roll is 2 or more yards less than that marked on ticket.
Roll contains more than four pieces.
Any piece in roll less than 20 yards in length.
Ends of pieces in roll not overlapped, ends joined by seam.

4.4.3 End item testing. Test results obtained under testing conditions as defined in FED-STD-191, FED-STD-601 or ASTM D 618 (Procedure A) will be acceptable except in case of dispute. In disputed cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

4.4.3.1 Physical testing. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. The physical values specified in section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. For resistance to leakage and blocking tests, there shall be no evidence of any specimen failing. The sample unit for testing shall be 4 continuous yards full width of the coated and laminated cloth. The lot size shall be expressed in units of linear yards, and the sample size (number of sample units) shall be as specified below. The lot shall be unacceptable if any sample unit fails to meet any specified requirement. All test reports shall contain the individual values used in expressing the final results.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE IV. Physical tests

Characteristic	Requirement reference	Test method
Base fabric:		
Weave	3.3.1	Visual
Construction	3.3.1	5050
Coated and laminated cloth:		
Weight	Table I	5041
Breaking strength:		
Initial	Table I	5100
After oven aging	Table I	5850 1/ and 5100

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TABLE IV. Physical tests (cont'd)

Characteristic	Requirement reference	Test method
Resistance to abrasion	Table I	5304 2/
Tearing strength	Table I	ASTM D 1424
Resistance to leakage:		
Initial	Table I	5516 3/
After cold crack	Table I	5874 4/ and 5516 3/
Resistance to blocking	Table I	5872

1/ Temperature at 180°F ± 2°F for seven days.

2/ Except as follows:

a. Five specimens from each of warp and filling directions shall be tested.
b. The back side of the finished coated and laminated cloth shall be abraded.
c. The specimens shall be abraded for 300 continuous cycles under a tension of 5 pounds and a load of 3 pounds. The abradant shall be 3M Company, 240 grit, V weight, aluminum oxide abrasive cloth or equal.
d. The width of the breaking strength specimen shall be as specified in Method 5102.

3/ The face side of the finished coated/laminated cloth shall contact the water.

4/ Conditioned for 4 hours at minus 40°F ± 5°F prior to testing.

4.4.3.2 Toxicological agent resistance testing. Every roll of the finished, coated and laminated cloth shall have a sample of 3 continuous yards, full width, removed and forwarded to the U.S. Army Armament Munitions and Chemical Command for testing in accordance with table V. Any test failure shall be cause for rejection of the roll from which the failed sample was taken.

TABLE V. End item toxicological agents resistance tests (see 6.5)

Characteristic	Requirement paragraph	Test method
Resistance to mustard (H)	Table I	4.4.3.2.1
Resistance to thickened GD	Table I	4.4.3.2.2

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4.4.3.2.1 Resistance to mustard (H) test. Resistance to permeation by mustard (H) shall be determined in accordance with the Liquid Agent Contamination/Vapor Penetration Method described in CRDC-SP-84010. The test shall be conducted with a chemical agent challenge of 5 grams per square meter applied in droplets on the specimen surface. Test specimens shall be obtained by quartering a 3 yard piece of the coated and laminated cloth and selecting one specimen from each quarter; three specimens shall be tested on the face side and one specimen on the back side. No test specimen shall be taken within 2 inches from the selvage edge. The cumulative chemical agent penetration of each specimen shall not exceed 4 micrograms per square centimeter in the time interval specified in table I. All of the specimens shall meet the time requirement of table I.

4.4.3.2.2 Resistance to thickened GD test. Resistance to permeation by thickened GD shall be determined in accordance with the Liquid Agent Contamination/Vapor Penetration Method described in CRDC-SP-84010. The test shall be conducted with a chemical agent challenge of 5 grams per square meter applied in droplets on the specimen surface. Test specimens shall be obtained by quartering a 3 yard piece of the coated and laminated cloth and selecting two specimens from each quarter; six specimens shall be tested on the face side and two specimens on the back side. No test specimen shall be taken within 2 inches of the selvage edge. The cumulative chemical agent penetration of each specimen shall not exceed 10 micrograms per square centimeter in the time interval specified in table I. All of the specimens shall meet the time requirement of table I.

4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The coated cloth, shall be put up and packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The coated cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with PPP-P-1136.

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6. NOTES

6.1 Intended use. The coated and laminated cloth is intended for use as the ground sheet component of the chemical warfare agent resistant patient wrap.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (See 3.1, 4.3, and 6.4).
c. Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).

6.3 Sample. For access to standard samples, address the contracting activity issuing the invitation for bids (see 3.2).

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.5 Toxicological tests. Because of the extreme dangers involved in handling toxicological warfare agents, the special equipment required in connection with their use, all document tests involving toxicological agents will be performed by the U.S. Army Armament Munitions and Chemical Command (EA), Chemical Testing Branch, Building E5100, Aberdeen Proving Ground, MD 21010-5423 or an authorized independent Testing Laboratory approved by the U.S. Army Armament Munitions and Chemical Command.

6.6 Subject term (key word) listing.

Chemical protective
Cloth coated and laminated
Patient wrap

Custodians.

Army - GL
Navy - NU
Air Force - 11

Preparing activity:

Army - GL

Project No. 8305-0200

Review activities:

Army - EA
Air Force - 82, 99
DLA - CT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44304	Cloth, Coated and Laminated, Chemical Protective

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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